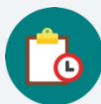
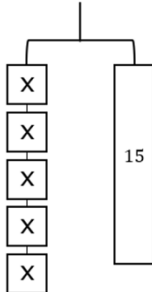
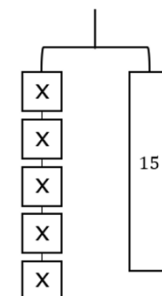


12.11 Solving One-Step Multiplication and Division Equations Using Diagrams

Lesson Introduction

 Benchmark	MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.		 Learning Target	I can solve one-step multiplication and division equations using diagrams.
 Benchmark Coherence	Building On MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- How do you use tape diagrams to model and solve one-step multiplication and division equations? - How do you use hanger diagrams to model and solve one-step multiplication and division equations? - How do you use number lines to model and solve one-step multiplication and division equations?			
 Lesson Narrative	In this lesson, students use manipulatives, tape diagrams, and number lines to solve multiplication and division one-step equations using inverse operations. This lesson begins with a hanger diagram in 12.11.2 Exploration where students use inverse operations on both sides of the equation to keep the equation balanced and find the solution. Students will then analyze tape diagrams to determine how to use them to solve equations and explore the relationships between equations and the number lines that model them with a partner. Students will then use tape diagrams, number lines, and hanger diagrams to write, model, and solve one-step multiplication and division equations.			
 Component Summary	12.11.1 Warm-Up (~5 min.)	Model and solve a one-step multiplication equation with algebra tiles (SE p. 340)	12.11.4 Collaboration (10-15 min.)	Match one-step multiplication and division equations and solutions (SE p. 344)
	12.11.2 Exploration (10 min.)	Explore how to model and solve one-step multiplication equations with hanger diagrams (SE p. 341)	12.11.5 Your Turn! (5 min.)	Practice using tape diagrams to model and solve one-step multiplication and division equations (SE p. 345)
	12.11.3 Guided Instruction (10 min.)	Use tape diagrams to model and solve one-step multiplication and division equations (SE p. 342)	12.11.6 Wrap-Up (~5 min.)	Use a visual model to solve a one-step multiplication equation (SE p. 345)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Algebra tiles - Colored pencils	- Consider creating an anchor chart with instructions, examples, and visual models. - Consider using virtual manipulatives.

 Teacher Tips	Common Misconceptions - Students may make computational errors when working with negative integers. - Students may not understand that the coefficient is multiplied by the variable because they do not see a multiplication symbol.	Things to Consider - In 12.11.2 Exploration, students are asked to provide positive feedback on their partner's work. Model some examples of positive feedback prior to the activity or provide students with sentence starters. - If time is an issue, then consider assigning 12.11.5 Your Turn! for homework.
	Literacy and Language Routines Take Turns In 12.11.4 Collaboration, students explain the rationale for their matches and listen to their partner's rationales, taking turns deciding, explaining, and listening. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students represent one-step multiplication problems with algebra tiles in the 12.11.1 Warm-Up. Then they represent one-step multiplication problems with a hanger diagram in 12.11.2 Exploration. Next, they use tape diagrams to represent one-step multiplication and division equations in 12.11.3 Guided Instruction.
	 Differentiate Instruction	
Hanger diagrams can be physical or digital manipulatives that benefit students who learn best with hands-on resources. Provide English Language Learners with anchors of support, such as vocabulary cards of the words used in this lesson: equation, positive, negative, inverse operations, integers, solutions. 		
Lesson Notes:		



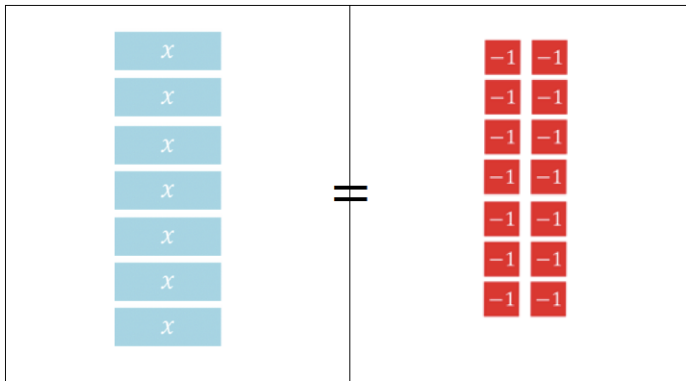
12.11.1 Warm-Up: Manipulative Math

Component Overview: Students model and solve a one-step multiplication equation using pictorial representations of algebra tiles.



12.11.1 Warm-Up: Manipulative Math

1. On the equation mat, sketch algebra tiles to represent the equation $7x = -14$, and show how to find the solution.



$$7x = -14$$

$$\frac{7x}{7} = \frac{-14}{7}$$

$$x = -2$$



Consider providing a kinesthetic entry point by giving students the option to use concrete algebra tiles.

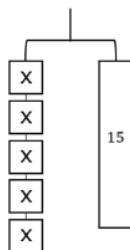
12.11.2 Exploration: Hanging Around Activity

Component Overview: Students explore how to model and solve one-step multiplication equations with hanger diagrams. Consider providing a hanger diagram template and having students trace square and rectangular shapes (i.e. algebra tiles) when sketching their diagrams.



12.11.2 Exploration: Hanging Around Activity

1. The hanger diagram represents the equation $5x = 15$.



- a. Complete the statements.

The left side of the hanger is broken into

- ☐ 3
- ☒ 5
- ☐ 15

equal parts. If 15 is broken into the same number of

equal parts, each part will have the same weight as

one x block. Splitting each side of the hanger into

- ☐ 3
- ☒ 5
- ☐ 15

the same as

- ☒ dividing
- ☐ multiplying

each

side of the equation by

- ☐ 3.
- ☒ 5.
- ☐ 15.

- b. Sketch splitting the hanger diagram into the equal parts as described in Part A.

Answers will vary.



This section provides a variety of entry points through the visual representations of the hanger diagrams and partner discussions. Consider providing algebra tiles for a kinesthetic entry point and encouraging students to model the equation first using tiles if they are struggling with the hanger diagram representation.

12.11.2 Exploration: Hanging Around Activity (continued)

- c. What is the value of x ?

$$x = 3$$

- d. Compare your diagram with your partner. Discuss the similarities and differences.

Answers will vary.

- e. Provide positive feedback to your partner in their book on their sketch and solution.

Answers will vary.



Model what positive feedback looks and sounds like prior to this activity with examples. Consider providing a posted anchor chart that students could reference.

Partner Feedback	
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12.11.3 Guided Instruction: Solving One-Step Multiplication and Division Equations Using Diagrams

Component Overview: Students work with a partner to model and solve one-step multiplication and division equations using tape diagrams. Consider having students trace square and rectangular shapes (i.e. algebra tiles) when sketching their diagrams.



12.11.3 Guided Instruction: Solving One-Step Multiplication and Division Equations Using Diagrams

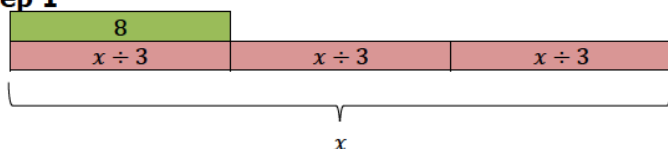
Tape diagrams, also known as bar models, can be used to model and solve multiplication and division equations.

- The equation $\frac{x}{3} = 8$ is represented by the tape diagram shown.

8
$x \div 3$

- The solution to the equation $\frac{x}{3} = 8$ can be modeled in the following steps. Work with your partner to explain what is occurring in each step.

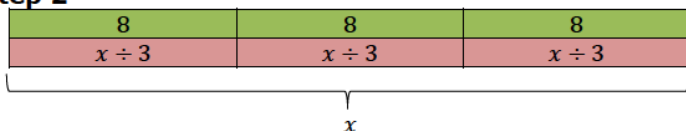
Step 1



Explanation of Step 1

Answers will vary. x is being broken into three sections. One section equals eight.

Step 2



Explanation of Step 2

Answers will vary. x is being broken into three equal sections. Since the sections are equal and the first section is equal to eight, section 2 and 3 are also equal to 8.



When introducing tape diagrams, ensure that students understand they represent part-to-whole relationships. Students may think 8 represents the whole because it is next to the equal sign. If you notice this misconception arising, ask questions like:

- Do we notice any patterns?
- x divided by 3 splits the whole x into how many parts? $-11x$ makes how many parts for $-x$?
- How many parts would the whole x have been split into if the question were " x divided by 4"?
- What do we notice about the total number of parts that are being created in relation to the numbers given? How can we use these parts to help us find the value of the variable?



If students are still confused about this relationship, it may be useful to label the parts and wholes on the tape diagrams.

12.11.3 Guided Instruction: Solving One-Step Multiplication and Division Equations Using Diagrams (continued)

Step 3

24
x

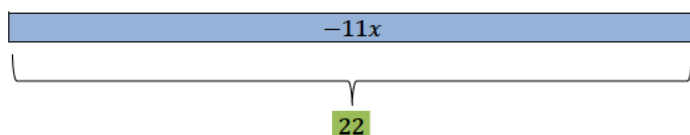
Explanation of Step 3

Answers will vary. Since each section equaled eight and there were three sections, x is equal to 24 because eight times three is 24 or $8 + 8 + 8 = 24$.

- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.

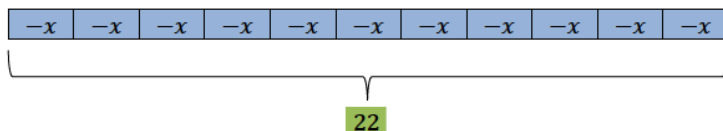
Answers will vary.

2. The tape diagram models the equation $22 = -11x$.



The solution to the equation $22 = -11x$ can be modeled in the following steps. Work with a different partner to explain what is occurring in each step.

Step 1



- a. How were the integers in the equation used to determine the negative value of x ?

Explanation of Step 1

Answers will vary. $-11x$ has been broken up into 11 individual pieces with each piece equivalent to $-x$.



Consider providing sentence stems for students to discuss their responses and explanations to provide additional entry points for learners.

"The operation used in the equation was _____. I solved the equation by using _____. This worked because..."

12.11.3 Guided Instruction: Solving One-Step Multiplication and Division Equations Using Diagrams (continued)

Step 2

$$\begin{array}{c} \boxed{-x} \\ \hline 2 \end{array}$$

Explanation of Step 2

Answers will vary. The value of $-x$ is 2.

Step 3

$$\begin{array}{c} \boxed{x} \\ \hline -2 \end{array}$$

Explanation of Step 3

Answers will vary. We want to find the value of x , so we take the opposite $-x$ and the opposite of 2, which gives us -2 .

- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.
Answers will vary.



If students are able to solve the equations without visual models, allow them to do so. Ask them to check their solutions. If they finish early, consider having them write real-world word problems that could be modeled by the equations in the lesson.



To encourage mathematical reasoning, ask questions like:

- *Does this answer make sense?*
- *Did you check your solution by substituting it back into the original equation?*
- *How can you use mental math to help you find the correct diagram that models the equation?*

12.11.4 Collaboration: Matching Equations and Solutions

Component Overview: Students match one-step multiplication and division equations and solutions. Then students explain how they determined which equations to match with which solutions to their partners.



12.11.4 Collaboration: Matching Equations and Solutions

1. Match each equation to the correct solution. Record the equation with its solution in the table. Some solutions may be used more than once or not at all. Include an explanation for the match.

A. $x = 2$	E. $x = 3$
B. $x = -20$	F. $x = 10$
C. $x = -10$	G. $x = -8$
D. $x = 9$	H. $x = -3$

1. $-3x = -6$
2. $-\frac{x}{2} = 4$
3. $\frac{x}{3} = 3$
4. $-\frac{x}{2} = 5$
5. $3x = 9$
6. $\frac{x}{10} = -2$
7. $4x = 8$



Take Turns

Students explain the rationale for their matches and listen to their partner's rationales, taking turns deciding, explaining, and listening. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.

12.11.4 Collaboration: Matching Equations and Solutions (continued)

Equation	Solution	Explanation for Match
1	A	Answers will vary. When substituting 2 in for x in $-3x = -6$ the result is -6 .
2	G	Answers will vary. When substituting -8 in for x in $-\frac{x}{2} = 4$ the result is 4.
3	D	Answers will vary. When substituting 9 in for x in $\frac{x}{3} = 3$ the result is 3.
4	C	Answers will vary. When substituting -10 in for x in $-\frac{x}{2} = 5$ the result is 5.
5	E	Answers will vary. When substituting 3 in for x in $3x = 9$ the result is 9.
6	B	Answers will vary. When substituting -20 in for x in $\frac{x}{10} = -2$ the result is -2 .
7	A	Answers will vary. When substituting 2 in for x in $4x = 8$ the result is 8.



Make manipulatives available to students who wish to use them to model and solve the equations. Also, remind students that they can draw visual models. This will provide additional visual and kinesthetic entry points.

12.11.5 Your Turn!

Component Overview: Students practice using tape diagrams to model and solve one-step multiplication and division equations.



12.11.5 Your Turn!

Sketch a tape diagram to represent each equation. Then, solve the equation.

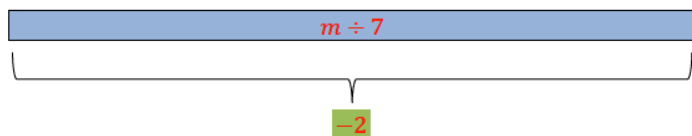
1. $-5p = -40$



$$\frac{-5p}{-5} = \frac{-40}{-5}$$

$$p = 8$$

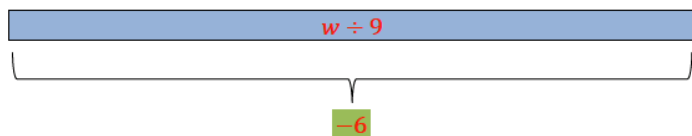
2. $-2 = \frac{m}{7}$



$$-2 \cdot 7 = \frac{m}{7} \cdot 7$$

$$m = -14$$

3. $-6 = \frac{w}{9}$



$$-6 \cdot 9 = \frac{w}{9} \cdot 9$$

$$w = -54$$



Display an anchor chart with instructions, examples, and visual models for struggling students or provide digital or physical manipulatives like hanger diagrams or algebra tiles for students to use to model each equation.

12.11.6 Wrap-Up: Ticket Out the Door

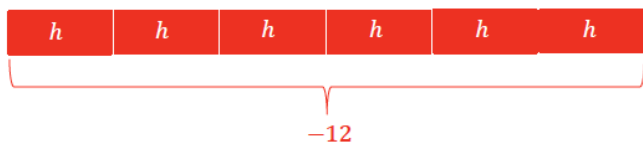
Component Overview: Students use a visual model to solve a one-step multiplication equation. Use data from this formative assessment to determine which students need remediation before the next lesson.



12.11.6 Wrap-Up: Ticket Out the Door

- Throughout this unit, you have been shown how to solve equations using tape diagrams, hanger diagrams, number lines, manipulatives, and inverse operations. Use one of these methods to model and solve the equation $6h = -12$.

Diagram answers will vary. $h = -2$



This Wrap-Up allows for multiple entry points, as students have a choice for how they want to represent and solve the equation. Encourage students to choose the method that makes the most sense to them.